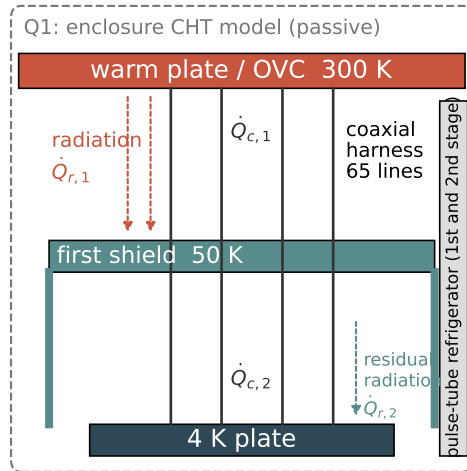


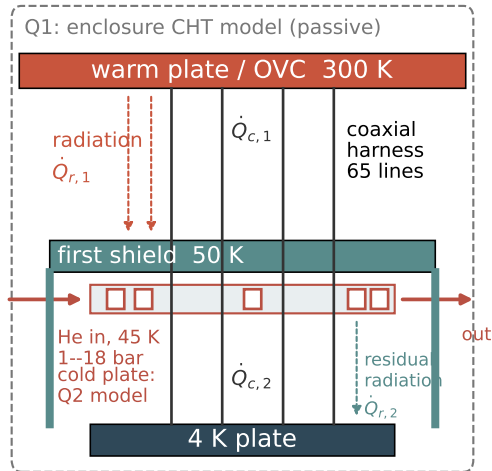
(a) reference: PTR first stage
40--50 K, conductive



dilution unit below 4 K: not modelled

duty removed by conduction to the PTR;
no coolant circuit

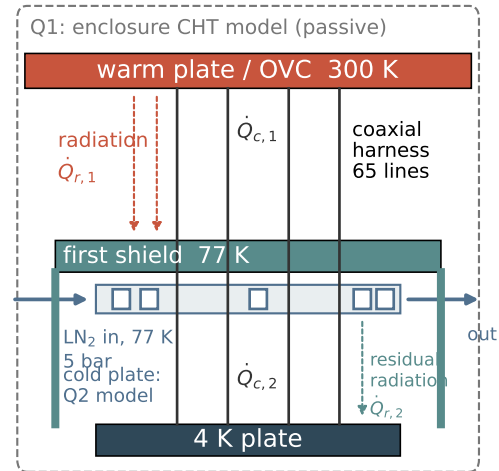
(b) route He: circulated gas
50 K nominal; 45 K in, 1--18 bar



dilution unit below 4 K: not modelled

duty $\dot{Q}_1$ removed by the coolant:
He cryoplant and circulator

(c) route N₂: liquid nitrogen
77 K in, 5 bar



dilution unit below 4 K: not modelled

duty $\dot{Q}_1$ removed by the coolant:
LN₂ dewar or pressurised loop